

Data Ingestion from the RDS to HDFS using Sqoop

Sqoop command used for importing table from RDS to HDFS

1. Command to import card_member table from RDS to hdfs system

```
sqoop import --connect "jdbc:mysql://upgradawsrds1.cyaie1c9bmnf.us-east-1.rds.amazonaws.com:3306/cred_financials_data?connectionCollation=latin1_swedish_ci" \  
--table card_member \  
--target-dir /user/hadoop/sqoop/import/cred_financials_data/card_member \  
--username upgraduser \  
-m 1 \  
--password upgraduser
```

2. Command to import member_score table from RDS to hdfs system

```
sqoop import --connect "jdbc:mysql://upgradawsrds1.cyaie1c9bmnf.us-east-1.rds.amazonaws.com:3306/cred_financials_data?connectionCollation=latin1_swedish_ci" \  
--table member_score \  
--target-dir /user/hadoop/sqoop/import/cred_financials_data/member_score \  
--username upgraduser \  
-m 1 \  
--password upgraduser
```

Command to see the list of imported data in HDFS

1 List the data command:

```
hadoop fs -ls /user/hadoop/sqoop/import/cred_financials_data/  
hadoop fs -ls  
/user/hadoop/sqoop/import/cred_financials_data/card_member/  
hadoop dfs -ls sqoop/import/cred_financials_data/member_score
```

2 Data validation command:

```
hadoop fs -cat
/user/hadoop/sqoop/import/cred_financials_data/card_member/part-m-
00000 | wc -l

hadoop fs -cat sqoop/import/cred_financials_data/member_score/part-m-
00000 | wc -l
```

Screenshot of the imported data along with count of each file.

- Output of table `card_member` import command with 999 rows.

```
[root@ip-172-31-4-189 ~]# sqoop import --connect 'jdbc:mysql://upgradawsdev1.cyaie1c9mmzf.us-east-1.rds.amazonaws.com:3306/cred_financials_data?connectionCollation=latin1_swedish_ci' --table card_member --target-dir /user2/hadoop/sqoop/in
port/cred_financials_data/card_member --username upgraduser -e 1 --password upgraduser
Warning: /usr/lib/sqoop/.accumulo does not exist! Accumulo imports will fail.
Please set HACCUMULO_HOME to the root of your Accumulo installation.
24/01/18 08:25:24 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/lib/sqoop/slf4j-impl-2.4.2.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html/multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
24/01/18 08:25:24 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
24/01/18 08:25:24 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
24/01/18 08:25:24 INFO tool.CodeGenTool: Beginning code generation
Loading class 'com.mysql.jdbc.Driver'. This is deprecated. The new driver class is 'com.mysql.cj.jdbc.Driver'. The driver is automatically registered via the SPI and manual loading of the driver class is generally unnecessary.
24/01/18 08:25:25 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM 'card_member' AS t LIMIT 1
24/01/18 08:25:25 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM 'card_member' AS t LIMIT 1
24/01/18 08:25:25 INFO org.apache.hadoop.mapred.lib.NativeDistributedCache: HADOOP_NATIVE_HOME is /usr/lib/hadoop-mapreduce
Note: /tmp/sqoop-root/compile/ca9b3ae2b150c2f66a9a782751a28/card_member.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
24/01/18 08:25:27 INFO org.apache.hadoop.mapred.lib.NativeDistributedCache: Writing jar file: /tmp/sqoop-root/compile/ca9b3ae2b150c2f66a9a782751a28/card_member.jar
24/01/18 08:25:27 WARN manager.MySQLManager: It looks like you are importing from mysql.
24/01/18 08:25:27 WARN manager.MySQLManager: This transfer can be faster! Use the --direct
24/01/18 08:25:27 WARN manager.MySQLManager: option to exercise a MySQL-specific fast path.
24/01/18 08:25:27 INFO manager.MySQLManager: Setting zero DATETIME behavior to convertToNull (mysql)
24/01/18 08:25:27 INFO mapreduce.ImportJobBase: Beginning import of card_member
24/01/18 08:25:27 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
24/01/18 08:25:28 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
24/01/18 08:25:28 INFO Configuration.deprecation: session.id is deprecated. Instead, use dfs.metrics.session-id
24/01/18 08:25:28 INFO yrm.YRMetrics: Initializing YRM Metrics with processName=jobtracker, sessionId=
24/01/18 08:25:28 INFO org.apache.hadoop.mapred.lib.NativeDistributedCache: Writing jar file: /tmp/sqoop-root/compile/ca9b3ae2b150c2f66a9a782751a28/card_member.jar
```

```
24/01/18 08:25:31 INFO mapred.Task: Final Counters for attempt_local1899944623_0001_m_000000_0: Counters: 15
File System Counters
  FILE: Number of bytes read=22390559
  FILE: Number of bytes written=23042119
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
Map-Reduce Framework
  Map input records=999
  Map output records=999
  Input split bytes=87
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=0
  Total committed heap usage (bytes)=459800576
File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=85757
24/01/18 08:25:31 INFO mapred.LocalJobRunner: Finishing task: attempt_local1899944623_0001_m_000000_0
24/01/18 08:25:31 INFO mapred.LocalJobRunner: map task executor complete.
24/01/18 08:25:31 INFO mapreduce.Job: Job job_local1899944623_0001 running in uber mode : false
24/01/18 08:25:31 INFO mapreduce.Job: map 100% reduce 0%
24/01/18 08:25:31 INFO mapreduce.Job: Job job_local1899944623_0001 completed successfully
24/01/18 08:25:31 INFO mapreduce.Job: Counters: 15
File System Counters
  FILE: Number of bytes read=22390559
  FILE: Number of bytes written=23042119
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
Map-Reduce Framework
  Map input records=999
  Map output records=999
  Input split bytes=87
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=0
  Total committed heap usage (bytes)=459800576
File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=85757
24/01/18 08:25:31 INFO mapreduce.ImportJobBase: Transferred 0 bytes in 3.7118 seconds (0 bytes/sec)
24/01/18 08:25:31 INFO mapreduce.ImportJobBase: Retrieved 999 records.
```

• Output of table member_score import command with 999 rows.

```
[root@ip-172-31-4-180 ~]# sqoop import --connect jdbc:mysql://sgw1awsrds1.cysatclmmtf.us-east-1.rds.amazonaws.com:3306/cred_financials_data --username upgraduser -H 1 --password upgraduser --import-credential --table member_score --target-dir /user/hadoop/sqoop/
Warning: Use of libsqoop.so.0: Accumulo does not exist: Accumulo imports will fail.
Please see $ACCUMULO_HOME to the root of your Accumulo installation.
24/01/18 08:32:05 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/share/awx/redashift-jdbc/redashift-jdbc42-1.2.37.1061.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-sql-2.6.2.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
24/01/18 08:32:05 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
24/01/18 08:32:05 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
24/01/18 08:32:05 INFO tool.CodeGenTool: Beginning code generation
Loading class 'com.mysql.jdbc.Driver'. This is deprecated. The new driver class is 'com.mysql.cj.jdbc.Driver'. The driver is automatically registered via the SPI and manual loading of the driver class is generally unnecessary.
24/01/18 08:32:06 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM 'member_score' AS t LIMIT 1
24/01/18 08:32:06 INFO org.CompileAnnotationManager: HADOOP MAPREDUCE is jar:/lib/hadoop-mapreduce
Note: /tmp/sqoop-root/compile/cfbd0f63rae7d60267489cf1b17b70b4/member_score.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
24/01/18 08:32:08 INFO org.CompileAnnotationManager: Writing jar file: /tmp/sqoop-root/compile/cfbd0f63rae7d60267489cf1b17b70b4/member_score.jar
24/01/18 08:32:08 WARN manager.MySQLManager: It looks like you are importing from mysql.
24/01/18 08:32:08 WARN manager.MySQLManager: This transfer can be faster! Use the --direct
24/01/18 08:32:08 WARN manager.MySQLManager: option to execute a MySQL-specific fast path.
24/01/18 08:32:08 INFO manager.MySQLManager: Setting zero DATETIME behavior to convertToNull (mysql)
24/01/18 08:32:08 INFO mapreduce.ImportJobBase: Beginning import of member_score
24/01/18 08:32:08 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
24/01/18 08:32:09 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
24/01/18 08:32:09 INFO Client.RMPProxy: Connecting to ResourceManager at ip-172-31-4-180.ec2.internal/172.31.4.180:8032
24/01/18 08:32:15 INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat: Using red committed transaction isolation
24/01/18 08:32:19 INFO mapreduce.JobSubmitter: number of splits=1
24/01/18 08:32:19 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1705566423041_0001
24/01/18 08:32:20 INFO impl.YarnClientImpl: Submitted application application_1705566423041_0001
24/01/18 08:32:20 INFO mapreduce.Job: The url to track the job: http://ip-172-31-4-180.ec2.internal:20888/proxy/application_1705566423041_0001/
24/01/18 08:32:20 INFO mapreduce.Job: Running Job: job_1705566423041_0001
24/01/18 08:32:27 INFO mapreduce.Job: Job job_1705566423041_0001 running in uber mode : false
24/01/18 08:32:27 INFO mapreduce.Job: map 0% reduce 0%
24/01/18 08:32:28 INFO mapreduce.Job: map 100% reduce 0%
24/01/18 08:32:35 INFO mapreduce.Job: Job job_1705566423041_0001 completed successfully
24/01/18 08:32:35 INFO mapreduce.Job: Counters: 30
File System Counters
  FILE: Number of bytes read=0
  FILE: Number of bytes written=185753
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
HDFS: Number of bytes read=0
```

```
Note: /tmp/sqoop-root/compile/cfbdf063fae7d60267489cf1b17bf0b4/member_score.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
24/01/18 08:32:08 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-root/compile/cfbdf063fae7d60267489cf1b17bf0b4/member_score.jar
24/01/18 08:32:08 WARN manager.MySQLManager: It looks like you are importing from mysql.
24/01/18 08:32:08 WARN manager.MySQLManager: This transfer can be faster! Use the --direct
24/01/18 08:32:08 WARN manager.MySQLManager: option to exercise a MySQL-specific fast path.
24/01/18 08:32:08 INFO manager.MySQLManager: Setting zero DATETIME behavior to convertToNull (mysql)
24/01/18 08:32:08 INFO mapreduce.ImportJobBase: Beginning import of member_score
24/01/18 08:32:08 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
24/01/18 08:32:09 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
24/01/18 08:32:09 INFO client.RMProxy: Connecting to ResourceManager at ip-172-31-4-180.ec2.internal/172.31.4.180:8032
24/01/18 08:32:19 INFO db.DBInputFormat: Using read committed transaction isolation
24/01/18 08:32:19 INFO mapreduce.JobSubmitter: number of splits=1
24/01/18 08:32:19 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1705566423041_0001
24/01/18 08:32:20 INFO impl.YarnClientImpl: Submitted application application_1705566423041_0001
24/01/18 08:32:20 INFO mapreduce.Job: The url to track the job: http://ip-172-31-4-180.ec2.internal:20888/proxy/application_1705566423041_0001/
24/01/18 08:32:20 INFO mapreduce.Job: Running job: job_1705566423041_0001
24/01/18 08:32:27 INFO mapreduce.Job: Job job_1705566423041_0001 running in uber mode : false
24/01/18 08:32:27 INFO mapreduce.Job: map 0% reduce 0%
24/01/18 08:32:34 INFO mapreduce.Job: map 100% reduce 0%
24/01/18 08:32:35 INFO mapreduce.Job: Job job_1705566423041_0001 completed successfully
24/01/18 08:32:35 INFO mapreduce.Job: Counters: 30
File System Counters
  FILE: Number of bytes read=0
  FILE: Number of bytes written=189753
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=87
  HDFS: Number of bytes written=19980
  HDFS: Number of read operations=4
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
Job Counters
  Launched map tasks=1
  Other local map tasks=1
  Total time spent by all maps in occupied slots (ms)=376608
  Total time spent by all reduces in occupied slots (ms)=0
  Total time spent by all map tasks (ms)=3923
  Total vcore-milliseconds taken by all map tasks=3923
  Total megabyte-milliseconds taken by all map tasks=12051456
Map-Reduce Framework
  Map input records=999
  Map output records=999
  Input split bytes=87
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=61
  CPU time spent (ms)=1700
  Physical memory (bytes) snapshot=306360320
  Virtual memory (bytes) snapshot=4613513216
  Total committed heap usage (bytes)=311951360
File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=19980
24/01/18 08:32:35 INFO mapreduce.ImportJobBase: Transferred 19.5117 KB in 26.531 seconds (753.0805 bytes/sec)
24/01/18 08:32:35 INFO mapreduce.ImportJobBase: Retrieved 999 records.
```

- Data Validation and view imported files in HDFS

List the files imported in hdfs:

```
[hadoop@ip-172-31-4-180 ~]$ hadoop fs -ls /user/hadoop/sqoop/import/cred_financials_data/card_member/
Found 2 items
-rw-r--r-- 1 root hadoop 0 2024-01-18 08:42 /user/hadoop/sqoop/import/cred_financials_data/card_member/_SUCCESS
-rw-r--r-- 1 root hadoop 85081 2024-01-18 08:42 /user/hadoop/sqoop/import/cred_financials_data/card_member/part-m-00000
[hadoop@ip-172-31-4-180 ~]$

[hadoop@ip-172-31-4-180 ~]$ hadoop dfs -ls sqoop/import/cred_financials_data/member_score
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.

Found 2 items
-rw-r--r-- 1 root hadoop 0 2024-01-18 08:32 sqoop/import/cred_financials_data/member_score/_SUCCESS
-rw-r--r-- 1 root hadoop 19980 2024-01-18 08:32 sqoop/import/cred_financials_data/member_score/part-m-00000
[hadoop@ip-172-31-4-180 ~]$
```

No of rows imported in each file: 999

```
[hadoop@ip-172-31-4-180 ~]$ hadoop fs -cat /user/hadoop/sqoop/import/cred_financials_data/card_member/part-m-00000 | wc -l
999
[hadoop@ip-172-31-4-180 ~]$ hadoop fs -cat sqoop/import/cred_financials_data/member_score/part-m-00000 | wc -l
999
[hadoop@ip-172-31-4-180 ~]$
```